

If your task is **Perform EDA on Sales Data + 5 Visualizations**, here are the common commands and expected outputs.

1. Load Dataset

Command

```
import pandas as pd
```

```
df = pd.read_csv("sales.csv")
```

```
print(df.head())
```

Output

	OrderID	Product	Region	Sales	Profit
0	1001	Laptop	West	50000	12000
1	1002	Mobile	East	30000	8000
2	1003	Tablet	South	20000	5000
3	1004	Laptop	North	45000	10000
4	1005	Printer	West	10000	2000

2. Check Shape

Command

```
print(df.shape)
```

Output

```
(5000, 8)
```

Meaning:

- 5000 Rows
 - 8 Columns
-

3. Dataset Information

Command

```
print(df.info())
```

Output

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 5000 entries
```

```
Data columns (total 8 columns):
```

```
OrderID    5000 non-null int64
```

```
Product    5000 non-null object
```

```
Region     5000 non-null object
```

```
Sales      5000 non-null float64
```

```
Profit     5000 non-null float64
```

```
Quantity   5000 non-null int64
```

4. Statistical Summary

Command

```
print(df.describe())
```

Output

	Sales	Profit
count	5000.0000	5000.0000
mean	2500.5000	350.4000
std	1800.2000	200.5000
min	100.0000	-50.0000
max	15000.0000	3000.0000

Insight

- Average Sales = ₹2500
 - Maximum Sales = ₹15000
-

5. Check Missing Values

Command

```
print(df.isnull().sum())
```

Output

OrderID 0

Product 0

Region 0

Sales 0

Profit 0

Quantity 0

dtype: int64

Insight

No missing values found.

6. Check Duplicates

Command

```
print(df.duplicated().sum())
```

Output

5

Insight

5 duplicate records found.

Visualization 1

Sales by Region

Command

```
import matplotlib.pyplot as plt
```

```
df.groupby('Region')['Sales'].sum().plot(kind='bar')
```

```
plt.title("Sales By Region")
```

```
plt.show()
```

Output Graph

West 

East 

North 

South 

Insight

West region generated highest sales.

Visualization 2

Monthly Sales Trend

Command

```
df['Order Date'] = pd.to_datetime(df['Order Date'])
```

```
monthly_sales = df.groupby(  
df['Order Date'].dt.month)['Sales'].sum()
```

```
monthly_sales.plot(kind='line')
```

```
plt.show()
```

Output

Line graph showing monthly sales.

Insight

November and December have peak sales.

Visualization 3

Top 10 Products

Command

```
top_products = df.groupby('Product')['Sales'].sum()
```

```
top_products.sort_values(ascending=False).head(10).plot(kind='bar')
```

```
plt.show()
```

Output



Insight

Laptop is the highest-selling product.

Visualization 4

Profit Distribution

Command

```
plt.hist(df['Profit'], bins=20)
```

```
plt.title("Profit Distribution")
```

```
plt.show()
```

Output

Histogram showing profit frequency.

Insight

Most orders generate moderate profit.

Visualization 5

Sales vs Profit

Command

```
plt.scatter(df['Sales'], df['Profit'])
```

```
plt.xlabel("Sales")
```

```
plt.ylabel("Profit")
```

```
plt.show()
```

Output

Scatter Plot

Profit

```
^
|
|  *
| *  *
| *  *
| *  *
+-----> Sales
```

Insight

Higher sales generally lead to higher profit.

● **INTERVIEW QUESTIONS**

If your project is Sales Data Analysis using Python (EDA + Visualization), these are common interview/viva questions with answers:

Basic Questions

1. What is EDA?

Answer:

EDA (Exploratory Data Analysis) is the process of analyzing data to understand its structure, identify patterns, detect anomalies, and gain insights before building reports or models.

2. Why do we perform EDA?

Answer:

- Understand the dataset
 - Find missing values
 - Detect duplicates
 - Identify trends and patterns
 - Support business decision-making
-

3. Which libraries did you use?

Answer:

Pandas

NumPy

Matplotlib

Optional:

Seaborn

4. Why did you use Pandas?

Answer:

Pandas is used for:

- Reading datasets
 - Data cleaning
 - Data manipulation
 - Data analysis
-

5. How did you load the dataset?

Answer:

```
df = pd.read_csv("sales.csv")
```

or

```
df = pd.read_excel("sales.xlsx")
```

Dataset Questions

6. How many rows and columns were present?

Answer:

The dataset contained 5000 rows and 8 columns.

7. How did you check dataset information?

Answer:

```
df.info()
```

This provides:

- Column names
 - Data types
 - Null values
-

8. How did you check missing values?

Answer:

`df.isnull().sum()`

9. How did you check duplicate records?

Answer:

`df.duplicated().sum()`

10. How did you remove duplicates?

Answer:

`df.drop_duplicates(inplace=True)`

Statistical Questions

11. What does describe() do?

Answer:

`df.describe()`

It provides:

- **Count**
 - **Mean**
 - **Standard Deviation**
 - **Minimum**
 - **Maximum**
 - **Quartiles**
-

12. What is Mean?

Answer:

Mean is the average value.

Formula:

Mean = Sum of Values / Number of Values

13. Difference between Mean and Median?

Answer:

Mean	Median
Average	Middle value
Affected by outliers	Not affected much

Visualization Questions

14. Why did you use a Bar Chart?

Answer:

To compare sales among different regions or products.

15. Why did you use a Line Chart?

Answer:

To show trends over time, such as monthly sales.

16. Why did you use a Histogram?

Answer:

To understand the distribution of profit values.

17. Why did you use a Scatter Plot?

Answer:

To analyze the relationship between Sales and Profit.

Project-Specific Questions

18. Which region had the highest sales?

Answer:

The West region generated the highest sales.

19. Which product was the best-selling?

Answer:

Laptop was the best-selling product.

20. Which months had peak sales?

Answer:

November and December showed the highest sales.

21. What business insights did you find?

Answer:

- **West region performs best.**
 - **Laptop is the top revenue-generating product.**
 - **Sales increase during year-end months.**
 - **Most transactions are profitable.**
-

22. What business recommendations would you give?

Answer:

- **Increase inventory before November and December.**
 - **Focus marketing on high-performing regions.**
 - **Promote best-selling products.**
 - **Improve low-profit products.**
-

Python Questions

23. Difference between loc and iloc?

Answer:

df.loc[]

Uses labels.

df.iloc[]

Uses index positions.

24. What is a DataFrame?

Answer:

A DataFrame is a two-dimensional table-like structure in Pandas consisting of rows and columns.

25. What is GroupBy?

Answer:

```
df.groupby('Region')['Sales'].sum()
```

It groups data based on a column and performs aggregate operations.

Advanced Questions

26. What is Correlation?

Answer:

Correlation measures the relationship between two variables.

- +1 → Strong Positive
 - 0 → No Relationship
 - -1 → Strong Negative
-

27. What is Data Cleaning?

Answer:

Data cleaning involves:

- Removing duplicates
 - Handling missing values
 - Correcting errors
 - Standardizing data
-

28. What challenges did you face?

Answer:

- Understanding dataset structure

- **Selecting appropriate visualizations**
 - **Interpreting business insights from graphs**
-

29. Why did you choose Python for this project?

Answer:

Python provides powerful libraries such as Pandas, NumPy, and Matplotlib, making data analysis and visualization efficient and easy.

30. Explain your project in one minute.

Answer:

"This project focuses on Sales Data Analysis using Python. I performed Exploratory Data Analysis to understand the dataset, checked for missing values and duplicates, and generated visualizations to identify trends. The analysis showed that the West region had the highest sales, Laptop was the top-selling product, and sales peaked during November and December. Based on these findings, I provided business recommendations to improve sales and profitability."

This 1-minute explanation is often enough to answer the common "Explain your project" question in a viva or interview.